

Ruturaj Dixit

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SUMMARY

Applied AI Engineer with M.S. in Data Science (Pace University, GPA 3.6) and 4+ years building and shipping production AI systems — agentic pipelines, RAG, LLM fine-tuning, ML classifiers, fraud detection, recommender systems, and full-stack AI applications. Strong Python, LangChain/LangGraph, Hugging Face, FastAPI, PostgreSQL/pgvector. Published researcher (BICA 2026). Open source contributor (OplInfer, PyPI v1.0.0).

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Bash

AI/ML: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, PEFT/LoRA, LangChain, LangGraph, ReAct agents

LLMs & GenAI: OpenAI GPT-4o, Claude, Gemini, AWS Bedrock, QLoRA fine-tuning, vLLM, LLM-as-a-judge, prompt engineering, RAG, structured output schemas

Backend & APIs: FastAPI, REST APIs, microservices, Docker, CI/CD, cloud-native architecture

Cloud & MLOps: AWS (SageMaker, Lambda, S3, EC2), Azure AI Foundry, PostgreSQL, pgvector, Snowflake, dbt, PySpark

Frontend & Viz: React, TypeScript, Streamlit, Tableau, Power BI, Plotly, Recharts

EXPERIENCE

Applied AI Engineer | *Virtual Dental Care, Inc.* · New York, NY

Jul 2026 – Present

- Designed and deployed AI-enabled agentic pipeline for clinical research intelligence — LangGraph multi-agent orchestration, RAG over PostgreSQL/pgvector, structured output schemas, human-in-the-loop review gates; owned full application lifecycle post-launch with real users.
- Built scalable microservices and REST APIs on AWS (Lambda, S3, EC2) integrating LLM outputs into internal clinical workflows; implemented LLM-as-a-judge evaluation framework measuring every prompt and model change before shipping — drove measurable retrieval quality improvements.
- Built signal classifiers and entity extraction/linking systems using prompting and LoRA fine-tuned models; curated training data from analyst-approved outputs and ran fine-tuning jobs on open-weight models; reviewed junior engineer deliverables and guided coding standards.

Graduate Teaching Assistant & AI Researcher | *Pace University* · New York, NY

Feb 2025 – May 2026

- Built and maintained AI research applications — RAG pipelines, signal classifiers, entity extraction/linking — owning full lifecycle from design through deployment; curated training data and ran LoRA fine-tuning on open-weight models.
- Translated research requirements into technical designs and functional specifications; mentored students on AI engineering, Python, and ML best practices; documented prompts, schemas, evaluation results, and operating procedures. Published at BICA 2026. GPA 3.6.

Data Scientist | *PrimeMarket AI* · New York, NY

Jul 2024 – Jun 2026

- Built AI-powered financial intelligence platform — LangGraph agentic orchestration, pgvector RAG, signal classifiers, entity extraction/linking, financial chart pattern recognition using image segmentation (OpenCV); owned production system post-launch for 1+ year.
- Designed FastAPI backend services and cloud-native architecture on AWS; applied AI fluency to automate analyst workflows with human review of model outputs; built and maintained evaluation sets measuring every model and prompt change before shipping.

Software Engineer – AI & Computer Vision | *Tata Consultancy Services* · Pune, India

Oct 2022 – Aug 2024

- Stellantis: Deployed computer vision defect detection pipeline (YOLO, Detectron2, OpenCV) at 90%+ accuracy integrated into manufacturing quality workflows via scalable Python REST APIs.
- Tokyo Electron: Built ML model for semiconductor process optimization achieving 89% prediction accuracy; automated engineer workflows with AI tooling and supported human review in high-stakes production environments.
- RBC: Built real-time financial analytics dashboards, portfolio risk ML models, and SQL/Python data pipelines; translated business requirements into technical designs and delivered against stakeholder priorities.
- Lamb Weston: Built demand forecasting and supply chain optimization models; owned production systems end-to-end; reviewed junior engineer work; documented application lifecycle practices across all four enterprise engagements.

PROJECTS

LLM QLoRA Fine-Tuning — ROUGE +32–68% — github.com/ruturajdixit99/llm-qlora-finetune

- Fine-tuned Qwen2.5-3B-Instruct on SAMSum dialogue summarization using 4-bit QLoRA (rank-16 LoRA, ~30M trainable params, <1% of model) on a single 8GB GPU. ROUGE-1: +32.4%, ROUGE-2: +67.7%, ROUGE-L: +41.5% vs. base model. Evaluation framework with before/after comparison on 50 held-out test dialogues.

Financial Transaction Risk Analyzer — 0.9957 AUC — github.com/ruturajdixit99/Financial-Transaction-Risk-Analyzer

- Simulated 1,000-customer x ~80-transaction dataset across 5 countries; engineered features from transaction aggregates (chargeback rate, high-risk merchant ratios, night transaction patterns); trained Random Forest (0.9957 AUC, 96% accuracy, 0.91 precision) + Isolation Forest unsupervised anomaly detection. Dual-model architecture for fraud/risk scoring at the customer level.

Multi-Agent Customer Support RAG — github.com/ruturajdixit99/multi-agent-customer-support-rag

- Cloud-native agentic application — LangGraph orchestration, pgvector RAG with cross-encoder reranking, structured output schemas, LLM-as-a-judge evaluation framework, human-in-the-loop review gates. Signal type and stance classifiers with LoRA fine-tuned open-weight models; entity extraction and linking; full lifecycle from design through operation with real users.

Trading Signals Dashboard — FastAPI + React/TypeScript — github.com/ruturajdixit99/trading-signals-dashboard

- Full-stack financial dashboard — FastAPI backend (yfinance, pandas) computing 20/50-day SMA golden/death cross + 14-day RSI buy/sell/hold signals with plain-language explanations; React/TypeScript + Recharts + Tailwind CSS frontend with live 15s quote polling, interactive price charts (1M/6M/1Y), and per-ticker caching. Production-style architecture with graceful partial failure handling.

Hybrid Movie Recommender — Precision@10: 0.64 — github.com/ruturajdixit99/movielens-hybrid-recommender

- Hybrid SVD collaborative filtering + TF-IDF content-based filtering with tunable alpha blend and cold-start popularity fallback on MovieLens (100K ratings, 9,700 movies); explainable recommendations (shared genres/tags rationale); RMSE 0.9882, Precision@10 0.6317, Recall@10 0.6200 at alpha=0.75. Interactive Streamlit dashboard and CLI.

Healthcare Agentic AI Scheduling Agent — github.com/ruturajdixit99/healthcare-agentic-ai

- AI scheduling agent with ReAct reasoning, microservices REST API, automated workflow execution with human review gates, HIPAA-conscious audit trail. Integrated AWS Lambda, Azure AI Foundry, LangChain tool use; evaluation sets measuring scheduling accuracy and hallucination rate before every model change shipped.

PUBLICATIONS

- Dixit R, Verma N, and Benjamin P. 2026. An Active Perception Pipeline for a Robot Cognitive Architecture. BICA 2026 — 17th Annual International Conference on Biologically Inspired Cognitive Architectures.
- Dixit R. 2025. Pattern Recognition in Financial Charts Using Image Segmentation Technique. Preprint. ResearchGate. researchgate.net/publication/389497552.

EDUCATION

M.S. Data Science — Pace University, New York, NY | GPA 3.6

May 2026

Relevant Coursework: Machine Learning, Deep Learning, NLP, Cloud Computing, Data Engineering, Applied Statistics, Database Systems, Software Engineering

B.Tech Electronics Engineering — Shivaji University

CERTIFICATIONS

- AWS Certified Machine Learning Engineer Associate
- Databricks Fundamentals Certified
- Neo4j Fundamentals — Neo4j, Aug 2026
- PCAP — Python Programming Essentials
- PySpark Essential Training — LinkedIn Learning, Jan 2026